

Roll No.

6671

B. C. A. (Second Semester)

EXAMINATION, 2025-26

Paper Fourth

STATISTICS

Course Code : B230204T

Time : Two Hours]

[Maximum Marks : 75

Note : This question paper is divided into three Sections—A, B and C. Attempt questions from all Sections as directed.

Inst. : The candidates are required to answer only in serial order. If there are many parts of a question, answer them in continuation.

Section—A

Very Short Answer Type Questions

Note : Attempt any *three* questions. Give answer of each question in about **50** words. Each question carries 3 marks. $3 \times 3 = 9$

1. (a) Find mean deviation for the following :

5, 7, 9, 11, 13

(b) Explain population and sample with example.

(c) The mean of 20 observations is 40. One value 50 was wrongly taken instead of 30. Find corrected mean.

(d) What is the difference between Deterministic and Non-deterministic experiments ?

(e) Find quartile deviation for the following data :

5, 7, 9, 12, 15, 18, 21

Section—B

Short Answer Type Questions

Note : Attempt any *four* questions. Answer each question in about **225** words. Each question carries 9 marks.

$$4 \times 9 = 36$$

2. (a) Explain primary and secondary data with example.

(b) Calculate Median for the following distribution :

X	f
10	4
15	6
20	8
25	5
30	2

(c) Define Dispersion. Explain the types of dispersion with example.

(d) What is Cumulative Frequency ? Make a less than C.F. table and more than C.F. table for the following data :

X	f
5	4
10	2
15	3
20	5
25	8
30	6

- (e) Calculate Mean for the following data using step deviation (assumed mean) method :

Class Interval	Frequency
0—10	5
10—20	8
20—30	12
30—40	10
40—50	5

- (f) The combined mean of two groups having 30 and 40 observations is 55. If the mean of first group is 50, find the mean of second group.
- (g) Find missing frequency f , if the median is 27.

C. I.	Frequency
0—10	4
10—20	6
20—30	f
30—40	10
40—50	5

Section—C

Long Answer Type Questions

Note : Attempt any *two* questions. Answer each question in about 475 words. Each question carries 15 marks.

$$2 \times 15 = 30$$

3. (a) Calculate standard deviation for the following grouped data :

Class Interval	Frequency
0—10	5
10—20	8
20—30	15
30—40	10
40—50	7

- (b) Define Event. What is the difference between Sample Space and Event ? Explain different types of events with examples.
- (c) A bag contains 8 black, 3 red and 9 white balls. If 3 balls are drawn at random, find the probability that :
- (i) All are black
 - (ii) 2 are black and 1 is red
 - (iii) One of each color.
- (d) From 8 men and 5 women a committee of 4 persons is to be formed. Find the number of ways in which the committee can be formed if it must contain :
- (i) Exactly 2 women
 - (ii) At least 1 woman.

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